Samuel Chan-ashing

PhD Researcher | Probability & Stochastic Analysis

Paris, France · samuel.chan-ashing@ens.psl.eu · Research webpage

Doctoral research

PhD in Mathematics, Université Paris Dauphine–PSL 2024–2027 (expected)

Probability Theory, Stochastic Analysis

Research affiliations: **ENS–PSL** (DMA) · **Inria** · **École Polytechnique** (CMAP).
Advisors: **Djalil Chafaï** and **Quentin Cormier**.
Thesis: *Cutoff and metastability phenomena for ergodic diffusion processes.*

Publications & research in progress

On the cutoff phenomenon for Dyson–Laguerre processes Published · 2026

S. Chan-ashing. *Stochastic Processes and their Applications*, 201:105054.

Derived explicit mixing estimates and cutoff using intrinsic Wasserstein distances associated with the intrinsic metric of the diffusion.

On the cutoff phenomenon for Dyson–Jacobi processes Submitted · 2026

S. Chan-ashing. *arXiv*

Extended cutoff analysis to Jacobi diffusions through a diffeomorphic change of variables, transferring spectral and curvature estimates between intrinsic and Euclidean formulations.

Exponentially long confinement near stable equilibria for mean-field diffusions Submitted · 2026

S. Chan-ashing, Q. Cormier. *arXiv*

Studying exponentially long confinement near stable McKean–Vlasov equilibria, including an explicit Ornstein–Uhlenbeck–cosine model.

Skills & methods

Programming:	Python (NumPy, SciPy, scikit-learn, Matplotlib); LaTeX.
Numerical methods:	stochastic simulation; Monte Carlo methods.
Mathematical methods:	stochastic analysis; Markov processes; optimal transport; spectral methods; functional inequalities; interacting particle systems; high-dimensional probability; random matrix theory.
Languages:	French and Réunionese Creole (native); English (professional working proficiency); German (basic).

National qualification

Agrégation in Mathematics, Probability and Statistics option 2022–2023

French national competitive examination for advanced teaching in mathematics.

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Research CV

Education

Sorbonne Université, Paris 2023–2024

MSc in Probability and Random Models.

École Normale Supérieure de Rennes 2020–2024

BSc with highest honors; first-year MSc in Pure Mathematics;
MSc preparation for the Agrégation with honors.

Research internships

DMA, École Normale Supérieure 3 months · 2024

Advisor: **Djalil Chafaï**.
Cutoff phenomena and convergence to equilibrium for ergodic diffusion processes.

DPMMS, University of Cambridge 2 months · 2022

Advisor: **Clément Mouhot**.
Random dynamical systems and mathematical questions related to turbulence.

CMLS, École Polytechnique 1 month · 2021

Advisor: **Samir Salem**.
Mean-field particle systems with Lipschitz interactions and the Vlasov equation.

Distinction & selected talks

Prix des Talents d'Outre-mer, CASODOM 2026

Award recognizing the achievements of overseas French talents.

Journée des Jeunes Probabilistes et Statisticiens 2025

Research presentation, Oléron, France.

ANR Conviviality workshop 2025

Research presentation, Université Lyon 1, France.

Teaching & leadership

Teaching Assistant, undergraduate mathematics 2024–present

Université Paris Dauphine–PSL: Introduction to Probability Theory; Analysis.

Mathematics oral examiner 2021–2024

IPESUP, Paris, and Lycée de l'Assomption, Rennes; preparatory classes.

President, Student Union of ENS Rennes 2021–2022

Administration, budgeting, legal representation and team coordination.